SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)

Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Problem Statement : AI-Powered Digital Twin Platform for Smart City Infrastructure

Industry Context

Smart cities increasingly adopt digital twin technology to simulate and monitor urban infrastructure in real time. AI, IoT, GIS, and cloud computing enable virtual replicas of roads, utilities, buildings, and transportation systems for predictive maintenance and strategic planning.

Problem Statement

Urban infrastructure management is challenging due to aging assets, increasing population,and limited real-time visibility. Traditional monitoring systems provide fragmented information and lack predictive capabilities. An AI-powered digital twin platform should create virtual models of city infrastructure by integrating real-time IoT data. The system should simulate infrastructure performance, detect anomalies, predict failures, and support data-driven urban planning. This solution will improve infrastructure reliability and maintenance efficiency.

Objectives

1. Create digital twins of city infrastructure.

2. Integrate real-time IoT data.

3. Predict infrastructure failures.

4. Monitor asset health continuously.

5. Support predictive maintenance.

6. Simulate urban scenarios.

7. Generate infrastructure performance reports.

8. Improve smart city planning.

Expected Outcomes

1. Enhanced infrastructure monitoring.

2. Reduced maintenance costs.

3. Improved asset reliability.

4. Better urban planning decisions.

5. Accurate predictive maintenance.

6. Increased operational efficiency.

7. Real-time infrastructure visualization.

8. Sustainable city development.

1. INTRODUCTION

The AI-Powered Digital Twin Platform for Smart City Infrastructure is designed to monitor and manage urban infrastructure using IoT, AI, GIS and cloud technologies.

The system creates virtual representations of infrastructure such as roads, bridges, buildings and utilities. IoT sensors provide real-time data, while AI analyses the data to detect abnormal conditions and predict possible failures.

Testing is performed to verify the correctness, reliability, security and usability of the proposed platform.

Objectives

1. Monitor infrastructure using real-time IoT data.
2. Create and update digital twins.
3. Detect abnormal infrastructure conditions.
4. Predict possible infrastructure failures.
5. Generate maintenance alerts.
6. Support smart city planning and reporting.

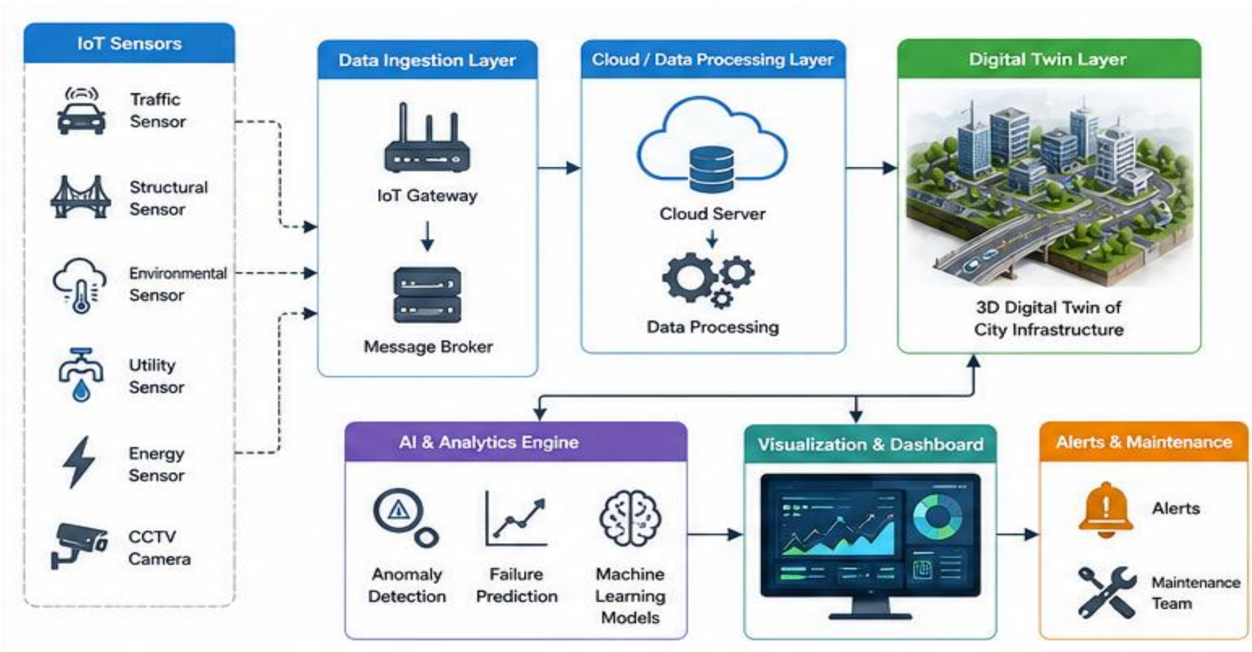

Figure 1: System Architecture

2. REQUIREMENTS TO BE TESTED

2.1 Functional Requirements

ID	Requirement
FR01	Register and manage infrastructure assets
FR02	Collect real-time IoT sensor data
FR03	Create and update digital twins
FR04	Display assets using GIS
FR05	Monitor asset health
FR06	Detect abnormal conditions
FR07	Predict infrastructure failures
FR08	Generate alerts
FR09	Recommend predictive maintenance
FR10	Generate infrastructure reports

2.2 Non-Functional Requirements

ID	Requirement
NFR01	Provide reliable infrastructure monitoring
NFR02	Provide acceptable response time
NFR03	Protect data from unauthorized access
NFR04	Support continuous operation
NFR05	Provide an easy-to-use dashboard
NFR06	Handle increasing sensor data

3. TEST SCENARIOS

ID	Test Scenario
TS01	Register infrastructure asset
TS02	Validate asset information
TS03	Receive IoT sensor data
TS04	Handle missing or invalid sensor data
TS05	Update digital twin
TS06	Display asset on GIS map
TS07	Monitor asset health
TS08	Detect abnormal sensor values
TS09	Predict infrastructure failure

ID	Test Scenario
TS10	Generate warning/critical alert
TS11	Generate maintenance recommendation
TS12	Generate infrastructure report
TS13	Authenticate users
TS14	Handle network or sensor failure

4. DETAILED TEST CASES

The test cases cover normal, boundary, invalid and exceptional conditions, as required by the assessment.

4.1 Asset and IoT Testing

ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC01	Register valid asset	Road RD001	Asset registered successfully	—	—
TC02	Missing asset ID	ID = Blank	Validation message displayed	—	—
TC03	Duplicate asset	RD001 existing	Duplicate registration rejected	—	—
TC04	Valid sensor data	Temperature = 30°C	Data displayed correctly	—	—
TC05	Invalid sensor data	Temperature = ABC	Invalid data rejected	—	—
TC06	Sensor disconnected	No data	Sensor failure detected	—	—
TC07	Boundary sensor value	Value = 100	Value processed correctly	—	—

4.2 Digital Twin and AI Testing

ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC08	Create digital twin	RD001	Digital twin created	—	—
TC09	Update digital twin	New sensor value	Twin updated	—	—

ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC10	Normal condition	Vibration = 2	Asset remains healthy	—	—
TC11	Abnormal condition	Vibration = 9.5	Anomaly detected	—	—
TC12	Critical condition	Vibration = 15	Critical alert generated	—	—
TC13	Failure prediction	Increasing vibration	Possible failure predicted	—	—
TC14	Insufficient data	One reading	Prediction not generated	—	—

Figure 2: Digital Twin Visualization

5. DASHBOARD, ALERT AND REPORT TESTING

ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC15	Display GIS location	Valid coordinates	Correct location displayed	—	—
TC16	Invalid coordinates	999,999	Input rejected	—	—
TC17	View asset health	RD001	Current health displayed	—	—

ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC18	Warning alert	Health = Warning	Warning generated	—	—
TC19	Critical alert	Health = Critical	Critical alert generated	—	—
TC20	Maintenance recommendation	High failure probability	Recommendation displayed	—	—
TC21	Generate report	Valid date range	Report generated	—	—
TC22	Empty report	No records	Appropriate message displayed	—	—
TC23	Invalid login	Wrong password	Access denied	—	—
TC24	Unauthorized access	Normal user → Admin	Access denied	—	—

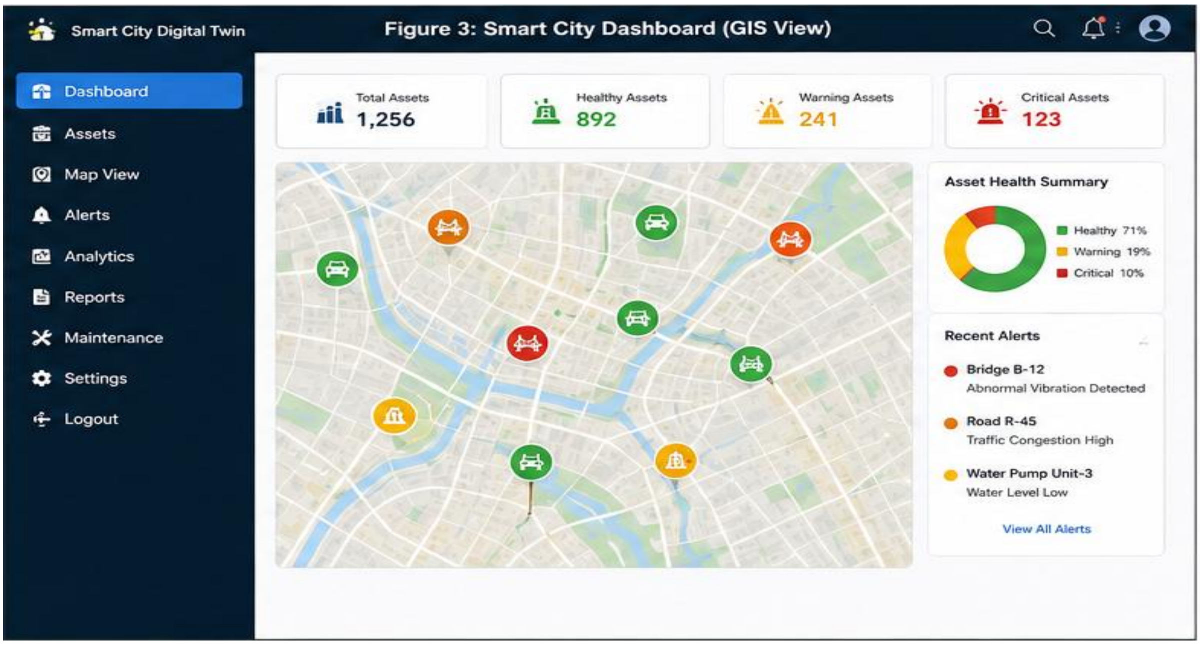

Figure 3: Smart City GIS Dashboard

6. TEST DESIGN TECHNIQUES

The assessment requires appropriate test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table and State Transition Testing.

6.1 Equivalence Partitioning

For an asset health score:

- 0–40 → Critical

- 41–70 → Warning
- 71–100 → Healthy
- Below 0 / Above 100 → Invalid

Input	Expected Result
-5	Invalid
20	Critical
55	Warning
90	Healthy
105	Invalid

6.2 Boundary Value Analysis

For the range 0–100, boundary values are tested.

Value	Expected Result
-1	Invalid
0	Critical
1	Critical
40	Critical
41	Warning
70	Warning
71	Healthy
100	Healthy
101	Invalid

6.3 Decision Table Testing

Sensor	Condition	Failure Risk	Expected Action
Connected	Normal	Low	No alert
Connected	Abnormal	Low	Warning

Sensor	Condition	Failure Risk	Expected Action
Connected	Abnormal	High	Critical alert
Disconnected	—	—	Sensor failure alert

6.4 State Transition Testing

The infrastructure asset follows these states:

Healthy → Warning → Critical → Maintenance → Healthy

- Normal readings → Healthy
- Abnormal readings → Warning
- Severe degradation → Critical
- Maintenance started → Maintenance
- Maintenance completed → Healthy

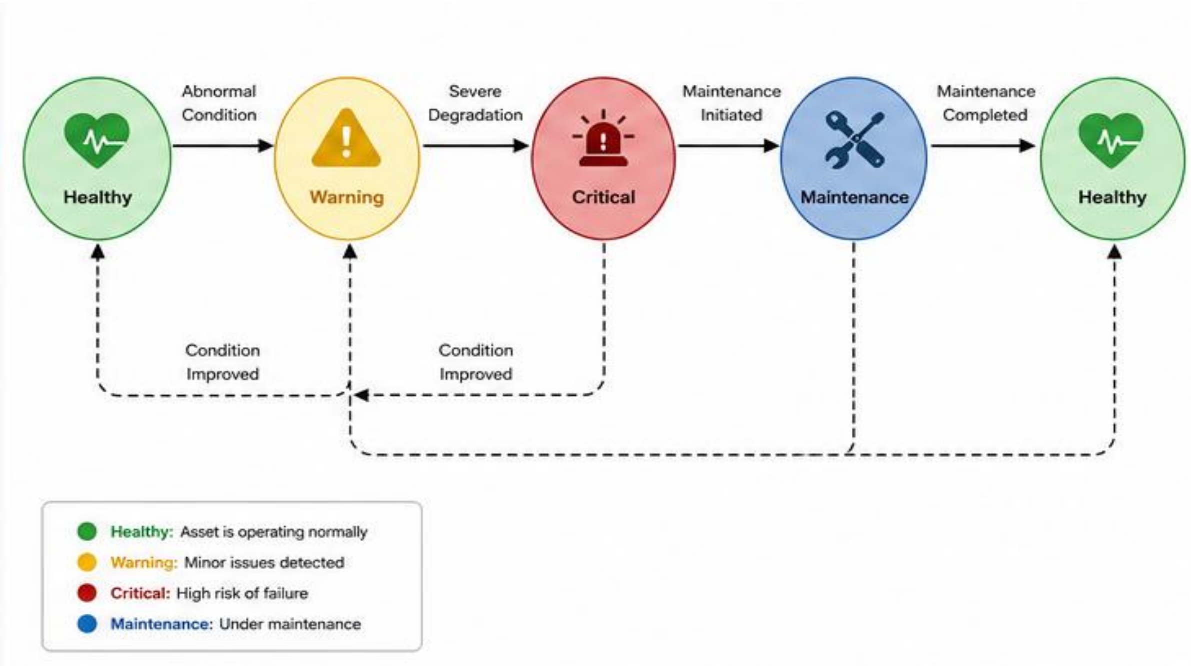

Figure 4: Asset Health State Transition

7. NON-FUNCTIONAL TESTING

ID	Test Type	Test Scenario	Expected Result
NFT01	Performance	Load dashboard with many assets	Dashboard remains responsive
NFT02	Reliability	Continuous monitoring	System operates without failure
NFT03	Security	Unauthorized login	Access denied
NFT04	Usability	New user operates dashboard	Functions are understandable
NFT05	Scalability	Increase sensor inputs	System handles increased load
NFT06	Recovery	Network failure	System recovers correctly

8. REQUIREMENT TRACEABILITY AND TEST COVERAGE

Requirement traceability maps requirements to test cases and ensures that the important requirements are tested. This directly addresses the assessment's Traceability & Test Coverage criterion.

Requirement	**Test Cases**
FR01 – Asset Management	TC01–TC03
FR02 – IoT Data	TC04–TC07
FR03 – Digital Twin	TC08–TC09
FR04 – GIS	TC15–TC16
FR05 – Asset Health	TC10, TC17
FR06 – Anomaly Detection	TC11–TC12
FR07 – Failure Prediction	TC13–TC14
FR08 – Alerts	TC18–TC19
FR09 – Maintenance	TC20
FR10 – Reports	TC21–TC22

Test Coverage

The proposed test cases cover:

- Functional requirements
- Non-functional requirements
- Normal conditions
- Boundary conditions

- Invalid inputs
- Exceptional conditions
- Security
- Failure and recovery

A total of 24 functional test cases and 6 non-functional test cases are proposed.

9. CONCLUSION

The proposed test case design verifies the major functions of the AI-Powered Digital Twin Platform for Smart City Infrastructure.

Testing covers IoT data collection, digital twin updates, GIS visualization, asset health monitoring, anomaly detection, AI-based prediction, alerts, predictive maintenance, reporting and security.

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing are applied to improve test coverage.

The requirement traceability matrix ensures that the major requirements are mapped to corresponding test cases. Therefore, the proposed testing approach helps ensure that the platform is reliable, secure and suitable for smart city infrastructure management.